

FCT SERVICE NOTE

SN05111710a

Class A Service information enclosed.

Date: November 17, 2005

Subject: VCT Air and Spectral Calibration Usage

Effectivity: VCT systems

Details:

M3A software came with two new calibration buttons on the Scanner Utilities screen. They are labeled as Air Calibration and Spectral Calibration. Misuse of these new calibration features can cause confusion due to incomplete calibrations.

These new cal selections may be used during troubleshooting but a complete pass of calibrations must be run. VCT calibrations must be run in a specific order and must be run to completion through each step.

The Air and Spectral calibration selections were intended primarily for manufacturing use but are available on the system, so please make sure you follow proper processes in using them.

Resolution:

The standard VCT calibration process is to run Collimator Cal, then Detailed Cal, then Fastcal. All 3 **MUST** be run back to back to properly update the calibration database. There are calibration vectors that Fastcal updates that Detailed cal does not as part of the VCT calibration design.

Air Calibration and Spectral Calibration are 2 subsets of Detailed Calibration. If you want to perform a spot calibration to try to troubleshoot an issue or quickly see if an issue is resolved due to hardware change or possible cal database issue, you can update a subset of the calibration database. This might save time if the issue was not resolved to continue troubleshooting, but if the issue was in the calibration database, a complete set of Detailed cals must be run followed by a Fastcal to completely update the calibration database after the spot calibration test prior to customer use.

To run a Spot calibration, an Air calibration can be run on just the small or large focal spot. This **must** then be followed by a Spectral calibration of the same focal spot, but you can also just choose a single FOV and slice collimation for Spectral calibration. This **must** then be followed by a FastCal to complete a spot calibration. Failure to run Fastcal after updating the air and spectral cal subsets will cause a mismatch in the calibration data due to the older Fastcal data being based on the older air and spectral cal data.

Problems have already been seen and misdiagnosed due to the use of spectral calibration only when the problem was actually in the air calibration database as well. A full pass of calibrations resolved the issue when a sole pass of spectral calibration did not. A spot calibration using Air cal, Spectral Cal followed by Fastcal, would have likely shown the issue to be fixed by recalibration. Using this procedure still would have required a full Detailed Cal and Fastcal to finish cleaning out the bad data in the cal database unless you knew for sure that the problem only existed in a known subset of the cal database.

Steve Merritt
FCT Service Engineering